

GEOLOGICAL FEATURES

Mars has two areas of markedly different terrain. Much of the northern hemisphere is characterized by relatively smooth and low-lying volcanic plains. The older southern landscape is typically cratered highland. The boundary between the two is an imaginary circle tilted by about 30° to the equator. The planet's major geological features are found within a region that extends roughly 30° each side of the equator. It contains Mars's main volcanic center, the Tharsis region, and the Valles Marineris, the vast canyon system that slices across the center of the planet.

WESTERN FLANK OF OLYMPUS MONS
Tectonic features on Mars take on familiar forms but are on a much grander scale than those on Earth. This escarpment on the side of Olympus Mons is 4.3 miles (7 km) high.

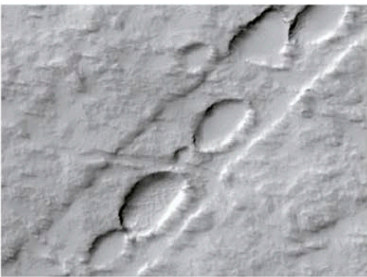
THARSIS MONTES

Pavonis Mons



TYPE Shield volcano
AGE 300 million years
DIAMETER 235 miles (375 km)

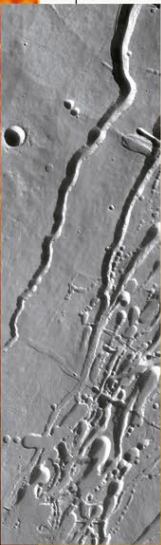
A huge bulge in the western hemisphere, commonly known as the Tharsis Bulge, contains volcanoes of various sizes and types, from large shields to smaller domes. Olympus Mons dominates the region. But three other volcanoes, which anywhere else would be considered enormous, are also found here. The three form a line and together make the Tharsis



PIT CHAIN
A chain of pits lies in a shallow trough on the lower east flank. The pits and trough formed because the ground was either moved apart by tectonic forces or uplifted by molten rock.

Montes mountain range. Pavonis Mons, situated on the equator, is the middle of the three. It is a shield volcano with a broad base and sloping sides and is similar to those found in Hawaii on Earth. The volcano's summit stands 4.3 miles (7 km) above the surrounding plain and has a single caldera within a larger, shallow depression. Hundreds of narrow lava flows are seen to emanate from the rim of the caldera, and others can be traced back to pits situated close by.

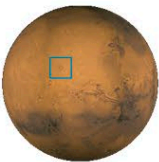
CHANNELS
These deep channels on the volcano's southern flank may have started out as subsurface lava tubes whose roofs collapsed as pits developed over them.



SUMMIT DEPRESSION
The summit caldera lies within a shallow depression that is almost twice the caldera's size and has faulted sides.

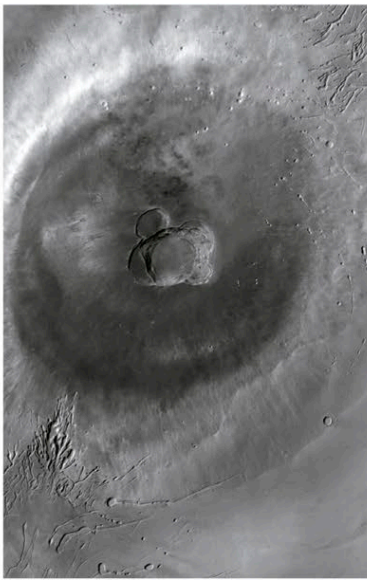
THARSIS MONTES

Ascræus Mons



TYPE Shield volcano
AGE 100 million years
DIAMETER 285 miles (460 km)

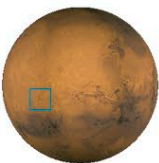
Ascræus Mons is the northernmost of the three Tharsis Montes volcanoes. The three lie on the crest of the Tharsis Bulge and form a line in a southwest–northeast direction. The line marks the position of a major rift zone, long since buried under lava. The three volcanoes grew by the gradual buildup of thousands of individual and successive lava flows that came to the surface through the rift zone. Ascræus is the tallest of the three, rising about 11 miles (18 km) above the surrounding plain. It has a large number of lines and channels all around the rim of the caldera, showing the paths taken by flowing lava.



CALDERA
The caldera on the summit is made up of eight major depressions and has a nested appearance (center). Its deepest point is over 1.9 miles (3 km) below the rim.

THARSIS MONTES

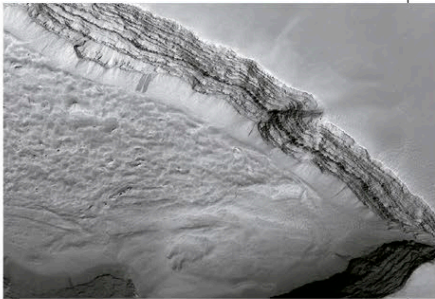
Arsia Mons



TYPE Shield volcano
AGE 700 million years
DIAMETER 295 miles (475 km)

Arsia Mons is second only to the mighty Olympus Mons in terms of volume. It is the southernmost of the three Tharsis Montes volcanoes, and its summit rises more than 5.6 miles (9 km) above the surrounding plain. Like the other two, it has a summit caldera bigger than any known on Earth. Arsia Mons measures 75 miles (120 km) across and is surrounded by arc-shaped faults. Lava flows fan out down the volcano's shallow slopes. The lava is of basaltlike composition and of low viscosity, and the flows are shorter nearer the summit than on the lower flanks.

CLOUDY SUMMIT
Water-ice clouds hang over the volcano's summit—a common sight every Martian afternoon in the Tharsis region.



LAYERED OUTCROP
This outcrop of layered rock lies in a pit on the volcano's lower west flank. The layers are thought to consist mostly of volcanic rock formed by successive lava flows.